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Simulation & Monte Carlo Methods

Project Report

Control Variates for a GARCH Model

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Introduction

Monte Carlo methods are widely used to approximate expectations and integrals that cannot be computed analytically. In Bayesian statistics, they are especially useful when posterior distributions are known only up to a normalising constant. This is the case in nonlinear time-series models such as GARCH models: the likelihood is recursive, the parameters are constrained by positivity and stationarity conditions, and the resulting posterior distribution is difficult to sample from directly. Markov Chain Monte Carlo (MCMC) methods provide a practical way to sample from such posteriors and estimate posterior quantities by empirical averages.

This project studies variance reduction for MCMC estimators using **Zero-Variance control variates**, following Mira, Solgi and Imparato (2013). The idea is to replace a target function f by a corrected function $\tilde{f}$ with the same posterior expectation but a smaller variance. The correction is built from the gradient of the log-posterior and a polynomial trial function, whose coefficients are estimated by regression on the MCMC output. The method can therefore be applied as a post-processing step.

The model considered is the Gaussian GARCH(1,1) model, a standard framework for financial returns with time-varying volatility and volatility clustering. The parameter vector $\omega = (\omega_1, \omega_2, \omega_3)$ satisfies $\omega_1 > 0$, $\omega_2, \omega_3 \geq 0$, and $\omega_2 + \omega_3 < 1$, making posterior simulation nontrivial and relevant for testing variance-reduction techniques.

The first part constructs a Random-Walk Metropolis sampler for the GARCH posterior. It is assessed on simulated data, where the true parameter is known, and then applied to real EUR/USD daily log-returns. The second part implements first-order Zero-Variance controls and compares ordinary MCMC with corrected estimators for ω_1 , ω_2 , ω_3 , and the persistence measure $\omega_2 + \omega_3$. The third part extends the control space to second-order polynomial controls and compares ZV-OLS, ZV-Lasso and ZV-Post-Lasso. Finally, the bonus part studies whether thinning or block averaging improves regression-based controls on autocorrelated MCMC chains.

The report is organised as follows. Chapter 1 presents the methodological framework. Chapter 2 assesses the MCMC sampler. Chapter 3 studies first-order control variates. Chapter 4 compares larger control spaces and regularised regression methods. Chapter 5 discusses autocorrelation in the MCMC chain.

Chapter 1

Methodological Framework

1.1 GARCH(1,1) Model and Posterior Distribution

The model

Let $r_1, \dots, r_T$ denote the observed return series, collected in the vector

$$r := (r_1, \dots, r_T) \in \mathbb{R}^T.$$

A Gaussian GARCH(1,1) model is considered:

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \quad (1.1)$$

where the conditional variance is recursively defined by

$$h_t = \omega_1 + \omega_2 r_{t-1}^2 + \omega_3 h_{t-1}, \quad t = 2, \dots, T. \quad (1.2)$$

The parameter vector is

$$\omega = (\omega_1, \omega_2, \omega_3)^\top.$$

The admissible region is defined as

$$\mathcal{A} = \{\omega \in \mathbb{R}^3 : \omega_1 > 0, \omega_2 \geq 0, \omega_3 \geq 0, \omega_2 + \omega_3 < 1\}. \quad (1.3)$$

The first three constraints ensure the positivity of the conditional variance, while the stationarity condition $\omega_2 + \omega_3 < 1$ guarantees a finite unconditional variance. Under this condition, the stationary initialization is chosen as

$$h_1 = \frac{\omega_1}{1 - \omega_2 - \omega_3}. \quad (1.4)$$

Prior distribution

Following the Bayesian setting used in the reference experiment, independent truncated Gaussian priors are assigned to the three GARCH parameters:

$$\omega_k \sim \mathcal{N}_{(0,+\infty)}(0, s_k^2), \quad k = 1, 2, 3, \quad s_k^2 = 100. \quad (1.5)$$

The positivity constraints are therefore incorporated in the prior support, while the additional stationarity restriction $\omega_2 + \omega_3 < 1$ is imposed during posterior evaluation. Any parameter vector outside $\mathcal{A}$ is assigned a log-posterior value equal to $-\infty$, and is consequently rejected by the Metropolis–Hastings algorithm.

Remark 1.1 (Justification of the prior).

- **Positivity constraints:** the truncation to the positive real line is consistent with the GARCH restrictions $\omega_1 > 0$, $\omega_2 \geq 0$, and $\omega_3 \geq 0$.

- **Diffuse specification:** the variance $s_k^2 = 100$ is large compared with the typical scale of GARCH parameters, which makes the prior weakly informative over the admissible region.
- **Gaussian shape:** this choice is consistent with the Bayesian GARCH experiment of Mira et al. (2013) and allows a direct comparison with their Zero-Variance MCMC setting. Moreover, the log-prior is quadratic and its gradient is linear, which simplifies the computation of the posterior score used to build the control variates. The light Gaussian tails also support the required integrability conditions for the ZV construction, although the equality $\mathbb{E}_\pi[\tilde{f}] = \mathbb{E}_\pi[f]$ ultimately depends on the regularity and boundary behavior of the full posterior density on the admissible region.
- **Independent components:** no prior dependence is assumed between ω_1 , ω_2 , and ω_3 .

Derivation of the log-posterior

Bayesian framework. By Bayes' rule, the posterior density of ω given the observed return series $r = (r_1, \dots, r_T)$ is

$$\pi(\omega \mid r) \propto \mathcal{L}(r \mid \omega) \pi_0(\omega), \quad (1.6)$$

or equivalently, in logarithmic form,

$$\log \pi(\omega \mid r) = \log \mathcal{L}(r \mid \omega) + \log \pi_0(\omega) + c, \quad (1.7)$$

where c denotes a constant independent of ω .

In the Metropolis–Hastings algorithm, the posterior is only required up to a multiplicative normalizing constant. Therefore, all terms independent of ω can be absorbed into c . The same applies to the normalizing constants of the likelihood and the truncated Gaussian prior.

Step 1 — Conditional structure. Given the past information

$$\mathcal{F}_{t-1} = \sigma(r_1, \dots, r_{t-1}),$$

the conditional distribution of r_t is Gaussian with variance h_t :

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t).$$

The variance h_t is a deterministic function of past observations and past conditional variances through the GARCH recursion.

Step 2 — Likelihood factorization. The joint likelihood can be written as a product of conditional densities:

$$\mathcal{L}(r \mid \omega) = \prod_{t=1}^T p(r_t \mid \mathcal{F}_{t-1}, \omega). \quad (1.8)$$

Since

$$p(r_t \mid \mathcal{F}_{t-1}, \omega) = \frac{1}{\sqrt{2\pi h_t}} \exp\left(-\frac{r_t^2}{2h_t}\right),$$

the likelihood becomes

$$\mathcal{L}(r \mid \omega) = \prod_{t=1}^T \frac{1}{\sqrt{2\pi h_t}} \exp\left(-\frac{r_t^2}{2h_t}\right). \quad (1.9)$$

Step 3 — Log-likelihood. Taking the logarithm of (1.9) gives

$$\log \mathcal{L}(r \mid \omega) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left[\log h_t + \frac{r_t^2}{h_t} \right]. \quad (1.10)$$

Since the first term does not depend on ω , it can be absorbed into the additive constant:

$$\log \mathcal{L}(r \mid \omega) = -\frac{1}{2} \sum_{t=1}^T \left[\log h_t + \frac{r_t^2}{h_t} \right] + \text{const.} \quad (1.11)$$

Step 4 — Log-prior. For independent truncated Gaussian priors with variance s_k^2 , the log-prior over the positive orthant is

$$\log \pi_0(\omega) = -\frac{1}{2} \sum_{k=1}^3 \frac{\omega_k^2}{s_k^2} + \text{const.} \quad (1.12)$$

Step 5 — Final log-posterior. Combining (1.11) and (1.12), the log-posterior is given, for $\omega \in \mathcal{A}$, by

$$\log \pi(\omega \mid r) = -\frac{1}{2} \sum_{t=1}^T \left[\log h_t + \frac{r_t^2}{h_t} \right] - \frac{1}{2} \sum_{k=1}^3 \frac{\omega_k^2}{s_k^2} + c. \quad (1.13)$$

For $\omega \notin \mathcal{A}$, the log-posterior is set to $-\infty$.

Gradient of the log-posterior

The analytical gradient of the log-posterior is required to construct the Zero-Variance control variates in the following sections. From (1.13), the prior contribution is immediate:

$$\frac{\partial}{\partial \omega_k} \log \pi_0(\omega) = -\frac{\omega_k}{s_k^2}, \quad k = 1, 2, 3. \quad (1.14)$$

The likelihood contribution follows from the chain rule. Since

$$\log \mathcal{L}(r \mid \omega) = -\frac{1}{2} \sum_{t=1}^T \left[\log h_t + \frac{r_t^2}{h_t} \right],$$

we obtain

$$\frac{\partial}{\partial \omega_k} \log \mathcal{L}(r \mid \omega) = -\frac{1}{2} \sum_{t=1}^T \left(\frac{1}{h_t} - \frac{r_t^2}{h_t^2} \right) \frac{\partial h_t}{\partial \omega_k}. \quad (1.15)$$

Equivalently,

$$\frac{\partial}{\partial \omega_k} \log \mathcal{L}(r \mid \omega) = -\frac{1}{2} \sum_{t=1}^T \frac{1}{h_t} \left(1 - \frac{r_t^2}{h_t} \right) \frac{\partial h_t}{\partial \omega_k}. \quad (1.16)$$

The derivatives of the conditional variance are computed recursively. For $t=2, \dots, T$

$$\frac{\partial h_t}{\partial \omega_1} = 1 + \omega_3 \frac{\partial h_{t-1}}{\partial \omega_1}, \quad (1.17)$$

$$\frac{\partial h_t}{\partial \omega_2} = r_{t-1}^2 + \omega_3 \frac{\partial h_{t-1}}{\partial \omega_2}, \quad (1.18)$$

$$\frac{\partial h_t}{\partial \omega_3} = h_{t-1} + \omega_3 \frac{\partial h_{t-1}}{\partial \omega_3}. \quad (1.19)$$

Using the stationary initialization $h_1 = \omega_1/(1 - \omega_2 - \omega_3)$, the initial derivatives are

$$\frac{\partial h_1}{\partial \omega_1} = \frac{1}{1 - \omega_2 - \omega_3}, \quad (1.20)$$

$$\frac{\partial h_1}{\partial \omega_2} = \frac{\omega_1}{(1 - \omega_2 - \omega_3)^2}, \quad (1.21)$$

$$\frac{\partial h_1}{\partial \omega_3} = \frac{\omega_1}{(1 - \omega_2 - \omega_3)^2}. \quad (1.22)$$

Therefore, for each $k = 1, 2, 3$, the log-posterior gradient is

$$\boxed{\frac{\partial}{\partial \omega_k} \log \pi(\omega \mid r) = -\frac{1}{2} \sum_{t=1}^T \frac{1}{h_t} \left(1 - \frac{r_t^2}{h_t}\right) \frac{\partial h_t}{\partial \omega_k} - \frac{\omega_k}{s_k^2}}. \quad (1.23)$$

This gradient is the key ingredient used later to construct the Zero-Variance control vector

$$z(\omega) = -\frac{1}{2} \nabla_{\omega} \log \pi(\omega \mid r).$$

1.2 MCMC Sampling from the Posterior

Why MCMC?

The objective. The posterior distribution derived in the previous section is used to compute expectations of the form

$$\mathbb{E}_{\pi}[\varphi(\omega)] = \int_{\mathcal{A}} \varphi(\omega) \pi(\omega \mid r) d\omega,$$

where φ may represent one of the GARCH parameters, the persistence $\omega_2 + \omega_3$, or a function involved in the construction of control variates. In particular, the Zero-Variance method later requires posterior moments and covariances involving the score-based control vector z , such as

$$\Sigma_{zz} = \mathbb{E}_{\pi}[(z - \mathbb{E}_{\pi}[z])(z - \mathbb{E}_{\pi}[z])^{\top}], \quad \sigma(z, f) = \mathbb{E}_{\pi}[(z - \mathbb{E}_{\pi}[z])(f - \mathbb{E}_{\pi}[f])],$$

where z denotes the posterior-score-based control vector.

The main obstacle is that the posterior density is only known up to a normalising constant:

$$\pi(\omega \mid r) = \frac{\tilde{\pi}(\omega)}{Z}, \quad \tilde{\pi}(\omega) = \mathcal{L}(r \mid \omega) \pi_0(\omega), \quad Z = \int_{\mathcal{A}} \tilde{\pi}(\omega) d\omega. \quad (1.24)$$

The normalising constant Z has no closed-form expression and would itself require evaluating a multidimensional integral over the constrained admissible region $\mathcal{A}$.

Why direct i.i.d. sampling is not convenient. Classical i.i.d. simulation methods are not well suited to this posterior.

- **Inverse-CDF method.** This approach would require explicit marginal or conditional cumulative distribution functions of the posterior distribution, together with their inverses. These quantities are not available here, because the posterior is known only up to the normalising constant.
- **Accept-reject sampling.** Accept-reject sampling would require a proposal density g that is easy to sample from and dominates the target density on the whole admissible region. For the normalised posterior, this means finding a finite constant M such that

$$\pi(\omega \mid r) \leq M g(\omega), \quad \omega \in \mathcal{A}.$$

The acceptance probability is then $1/M$ on average. In practice, however, the posterior is only available through the unnormalised density $\tilde{\pi}(\omega) = L(r \mid \omega)\pi_0(\omega)$, so one would instead need a usable global envelope for $\tilde{\pi}$:

$$\tilde{\pi}(\omega) \leq Cg(\omega), \quad \omega \in A.$$

The resulting average acceptance probability would be Z/C , where Z is unknown. More importantly, constructing a tight envelope is difficult for this constrained GARCH posterior, since the likelihood may be highly concentrated and the support is restricted by positivity and stationarity conditions. A loose envelope would lead to many rejected samples, making the method inefficient.

The MCMC idea. Markov Chain Monte Carlo avoids the need for direct i.i.d. sampling. Instead of sampling independently from π , we construct a Markov chain $(\omega^{(0)}, \omega^{(1)}, \dots)$ whose stationary distribution is the posterior $\pi(\omega \mid r)$. The key advantage is that Metropolis–Hastings only uses ratios of unnormalised posterior densities:

$$\frac{\tilde{\pi}(\omega')}{\tilde{\pi}(\omega)} = \frac{\pi(\omega' \mid r)}{\pi(\omega \mid r)}.$$

The unknown normalising constant Z therefore cancels out.

Once the chain has reached stationarity, posterior expectations are estimated by ergodic averages:

$$\widehat{\mathbb{E}}_{\pi}[\varphi(\omega)] = \frac{1}{N} \sum_{i=1}^N \varphi(\omega^{(i)}). \quad (1.25)$$

Under standard ergodicity assumptions,

$$\frac{1}{N} \sum_{i=1}^N \varphi(\omega^{(i)}) \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \mathbb{E}_{\pi}[\varphi(\omega)].$$

The Metropolis–Hastings Algorithm

The Metropolis–Hastings algorithm constructs a π -invariant Markov chain by combining a proposal kernel with an accept-reject correction. Starting from the current state ω , a candidate ω' is generated from a proposal density $q(\omega' \mid \omega)$. The general Metropolis–Hastings acceptance probability is

$$\alpha(\omega, \omega') = \min \left\{ 1, \frac{\tilde{\pi}(\omega')q(\omega \mid \omega')}{\tilde{\pi}(\omega)q(\omega' \mid \omega)} \right\}. \quad (1.26)$$

The transition kernel generated by this accept-reject mechanism is reversible with respect to π , meaning that it satisfies the detailed balance condition

$$\pi(\omega)P(\omega, d\omega') = \pi(\omega')P(\omega', d\omega).$$

Detailed balance implies that π is invariant for the chain. Therefore, if $\omega^{(i)} \sim \pi$, then $\omega^{(i+1)} \sim \pi$.

Random Walk Metropolis

Symmetric Gaussian proposal. In this project, a Random Walk Metropolis sampler is used. The proposal is a Gaussian perturbation of the current state:

$$\omega' = \omega^{(i)} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Sigma_{\text{prop}}). \quad (1.27)$$

The proposal density is

$$q(\omega' \mid \omega^{(i)}) = \varphi_{\Sigma_{\text{prop}}}(\omega' - \omega^{(i)}),$$

and is symmetric:

$$q(\omega' \mid \omega) = q(\omega \mid \omega').$$

Consequently, the proposal terms cancel in (1.26), and the acceptance probability reduces to

$$\alpha(\omega^{(i)}, \omega') = \min \left\{ 1, \frac{\tilde{\pi}(\omega')}{\tilde{\pi}(\omega^{(i)})} \right\}. \quad (1.28)$$

In practice, the computation is performed on the log scale. If

$$\ell(\omega) = \log \tilde{\pi}(\omega),$$

then

$$\log \alpha = \min \{ 0, \ell(\omega') - \ell(\omega^{(i)}) \}. \quad (1.29)$$

The proposal is accepted if

$$\log U < \log \alpha, \quad U \sim \mathcal{U}(0, 1).$$

This log-scale implementation avoids numerical overflow or underflow when the posterior density is very small.

Support constraints. The Gaussian proposal can generate candidates outside the admissible region

$$\mathcal{A} = \{ \omega_1 > 0, \omega_2 \geq 0, \omega_3 \geq 0, \omega_2 + \omega_3 < 1 \}.$$

No explicit truncation of the Gaussian proposal is required. Instead, the unnormalised posterior is defined as

$$\tilde{\pi}(\omega) = 0, \quad \omega \notin \mathcal{A}, \quad \text{or equivalently,} \quad \ell(\omega) = -\infty, \quad \omega \notin \mathcal{A}.$$

Therefore, any proposed value outside $\mathcal{A}$ is automatically rejected.

Markov property and time-homogeneity. The sequence $(\omega^{(i)})_{i \geq 0}$ is a Markov chain because the next state depends only on the current state and on newly generated random variables. More precisely, given $\omega^{(i)}$, the algorithm draws

$$\varepsilon_{i+1} \sim \mathcal{N}(0, \Sigma_{\text{prop}}) \quad \text{and} \quad U_{i+1} \sim \mathcal{U}(0, 1),$$

independently of the past. The next state $\omega^{(i+1)}$ is then a deterministic function of

$$(\omega^{(i)}, \varepsilon_{i+1}, U_{i+1}).$$

Thus, the previous states

$$\omega^{(0)}, \dots, \omega^{(i-1)}$$

are not needed once $\omega^{(i)}$ is known.

The transition kernel is the same at every iteration, so the chain is time-homogeneous. For any measurable set $B \subset \mathcal{A}$, it can be written as

$$P(\omega, B) = \int_B \alpha(\omega, \omega') q(\omega' \mid \omega) d\omega' + \rho(\omega) \mathbf{1}_{\{\omega \in B\}}, \quad (1.30)$$

where

$$\rho(\omega) = 1 - \int_{\mathbb{R}^3} \alpha(\omega, \omega') q(\omega' \mid \omega) d\omega' \quad (1.31)$$

is the total rejection probability from ω .

Detailed balance. For the accepted-move part of the kernel, the symmetry of the proposal and the definition of α imply

$$\pi(\omega)\alpha(\omega, \omega')q(\omega' | \omega) = \pi(\omega')\alpha(\omega', \omega)q(\omega | \omega').$$

The rejection part only places probability mass on the diagonal $\omega' = \omega$, and is therefore automatically reversible. Hence the full transition kernel satisfies detailed balance:

$$\pi(\omega)P(\omega, d\omega') = \pi(\omega')P(\omega', d\omega).$$

As a consequence, π is an invariant distribution of the chain:

$$\int_{\mathcal{A}} \pi(d\omega)P(\omega, B) = \pi(B), \quad B \subset \mathcal{A}. \quad (1.32)$$

Proposal tuning. The efficiency of a Random Walk Metropolis sampler depends strongly on the proposal covariance Σ_{prop} . If the proposal variance is too small, the chain accepts many moves but explores the posterior slowly. If it is too large, most proposals are rejected and the chain remains stuck for long periods.

A standard tuning rule is the Roberts–Gelman–Gilks scaling:

$$\Sigma_{\text{prop}} = \frac{2.38^2}{d} \hat{\Sigma}_{\pi}, \quad (1.33)$$

where $d = 3$ is the parameter dimension and $\hat{\Sigma}_{\pi}$ is an estimate of the posterior covariance. This rule is motivated by asymptotic optimal scaling results for Random Walk Metropolis algorithms and targets an acceptance rate close to 0.234 in high-dimensional Gaussian settings. In the present finite-dimensional problem, the value 0.234 is used only as a guideline. The proposal covariance is estimated from a short pilot chain, and the final acceptance rate is checked empirically.

Asymptotic Properties and Practical Diagnostics

Ergodicity. Under standard regularity assumptions, the Random Walk Metropolis chain is π -irreducible, aperiodic, and has π as invariant distribution. These properties justify the use of ergodic averages. For any π -integrable function φ ,

$$\frac{1}{N} \sum_{i=1}^N \varphi(\omega^{(i)}) \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \mathbb{E}_{\pi}[\varphi(\omega)]. \quad (1.34)$$

This result plays the role of a law of large numbers for MCMC. The important difference with ordinary Monte Carlo is that the draws are not independent: successive values of the chain are correlated.

Burn-in. The first iterations of the chain may depend strongly on the arbitrary starting point $\omega^{(0)}$. Therefore, an initial burn-in period is discarded before computing posterior summaries. The remaining draws are treated as samples from the stationary regime.

Monte Carlo error and dependence. When a Markov chain central limit theorem applies, the MCMC estimator satisfies

$$\sqrt{N} \left(\frac{1}{N} \sum_{i=1}^N \varphi(\omega^{(i)}) - \mathbb{E}_{\pi}[\varphi(\omega)] \right) \implies \mathcal{N}(0, \sigma_{\varphi, \text{MCMC}}^2),$$

where

$$\sigma_{\varphi, \text{MCMC}}^2 = \gamma_0 + 2 \sum_{\ell=1}^{\infty} \gamma_{\ell}, \quad \gamma_{\ell} = \text{Cov}_{\pi}(\varphi(\omega^{(0)}), \varphi(\omega^{(\ell)})).$$

This variance includes the autocorrelation of the chain. It is generally larger than the variance that would be obtained from independent samples.

Effective sample size. The dependence between MCMC draws is summarised by the effective sample size:

$$ESS = \frac{N}{1 + 2 \sum_{\ell=1}^{\infty} \rho_{\ell}},$$

where ρ_{ℓ} is the lag- ℓ autocorrelation of the sampled quantity. A large ESS indicates that the chain provides many effectively independent draws.

Notation. Throughout the report,

$$(\omega^{(i)})_{i=1}^N \subset \mathcal{A}, \quad \omega^{(i)} = (\omega_1^{(i)}, \omega_2^{(i)}, \omega_3^{(i)}),$$

denotes the post-burn-in MCMC sample. Every posterior expectation required in the Zero-Variance construction is approximated by an ergodic average of the form (1.25).

1.3 Zero-Variance Principle

Goal. Let $f : \mathcal{A} \rightarrow \mathbb{R}$ be the function whose posterior expectation we want to compute:

$$\mu_f = \mathbb{E}_{\pi}[f(\omega)].$$

In the present project, typical choices of f are

$$f(\omega) = \omega_1, \quad f(\omega) = \omega_2, \quad f(\omega) = \omega_3, \quad f(\omega) = \omega_2 + \omega_3.$$

The Zero-Variance idea of Mira et al. [1] is to replace f by a modified function $\tilde{f}$ such that

$$\mathbb{E}_{\pi}[\tilde{f}] = \mathbb{E}_{\pi}[f],$$

but with a smaller variance:

$$\text{Var}_{\pi}(\tilde{f}) \ll \text{Var}_{\pi}(f).$$

The same MCMC sample $\omega^{(1)}, \dots, \omega^{(N)}$ can then be used to compute a more accurate estimator:

$$\hat{\mu}_{\text{ZV}} = \frac{1}{N} \sum_{i=1}^N \tilde{f}(\omega^{(i)}).$$

Main intuition. The original MCMC estimator averages $f(\omega^{(i)})$. If a zero-mean function $h(\omega)$ is strongly correlated with $f(\omega)$, then

$$f(\omega) + h(\omega)$$

has the same expectation as $f(\omega)$, but may have a much smaller variance. The Zero-Variance method provides a systematic way to construct such control variates from the derivatives of the log-posterior density.

Mathematical Objects

Before stating the Zero-Variance formula, we introduce the differential operators involved.

- **Gradient.** For a smooth function $g : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\nabla g(x) = \left(\frac{\partial g}{\partial x_1}(x), \dots, \frac{\partial g}{\partial x_d}(x) \right)^{\top}.$$

- **Laplacian.** The Laplacian is the trace of the Hessian:

$$\Delta g(x) = \sum_{j=1}^d \frac{\partial^2 g}{\partial x_j^2}(x).$$

- **Trial function.** A trial function is chosen in the form

$$\psi(x) = P(x)\sqrt{\pi(x)},$$

where P is a polynomial. In practice, we use low-degree polynomials, especially degrees 1 and 2.

- **Hamiltonian operator.** The Schrödinger-type Hamiltonian used by Mira et al. is

$$(Hg)(x) = -\frac{1}{2}\Delta g(x) + V(x)g(x),$$

with potential

$$V(x) = \frac{1}{2\sqrt{\pi(x)}}\Delta\sqrt{\pi(x)}.$$

Remark 1.2 (Design condition). *The potential V is chosen so that*

$$H\sqrt{\pi} = 0.$$

Indeed,

$$H\sqrt{\pi} = -\frac{1}{2}\Delta\sqrt{\pi} + \frac{1}{2\sqrt{\pi}}\Delta\sqrt{\pi}\sqrt{\pi} = 0.$$

This condition is the starting point of the Zero-Variance construction.

The Central Zero-Variance Formula

The renormalised function is defined by

$$\tilde{f}(x) = f(x) + \frac{H\psi(x)}{\sqrt{\pi(x)}}.$$

Using $\psi = P\sqrt{\pi}$, Mira et al. obtain the practical formula

$$\boxed{\tilde{f}(x) = f(x) - \frac{1}{2}\Delta P(x) + \nabla P(x)^\top z(x), \quad z(x) = -\frac{1}{2}\nabla \log \pi(x).} \quad (1.35)$$

This is the formula used throughout the project.

Interpretation of the terms.

- $f(x)$ is the original function to be integrated.
- The term $-\frac{1}{2}\Delta P(x)$ is a deterministic correction determined by the chosen polynomial P .
- The term $\nabla P(x)^\top z(x)$ is the actual control-variate correction.
- The vector

$$z(x) = -\frac{1}{2}\nabla \log \pi(x)$$

is based on the posterior score.

In the Bayesian setting, the posterior density is known only up to a normalising constant:

$$\pi(x) = \frac{\tilde{\pi}(x)}{Z}.$$

However,

$$\nabla \log \pi(x) = \nabla \log \tilde{\pi}(x),$$

because $\log Z$ is constant with respect to x . Therefore, the Zero-Variance controls can be computed without knowing the normalising constant of the posterior.

Remark 1.3 (Unbiasedness and boundary conditions). *The equality*

$$\mathbb{E}_\pi[\tilde{f}] = \mathbb{E}_\pi[f]$$

is not automatic for every posterior density and every polynomial P . It depends on regularity and boundary conditions. Mira et al. show that, using integration by parts,

$$\mathbb{E}_\pi \left[\frac{H\psi}{\sqrt{\pi}} \right] = \frac{1}{2} \int_{\partial\Omega} (\psi \nabla \sqrt{\pi} - \sqrt{\pi} \nabla \psi) \cdot n \, d\sigma, \quad (1.36)$$

where Ω is the support of the posterior density and n is the outward unit normal vector. If the boundary term vanishes, then

$$\mathbb{E}_\pi \left[\frac{H\psi}{\sqrt{\pi}} \right] = 0,$$

and consequently

$$\mathbb{E}_\pi[\tilde{f}] = \mathbb{E}_\pi[f].$$

In our GARCH application, the posterior support is the constrained admissible region

$$\Omega = \mathcal{A}.$$

Therefore, the boundary condition is the main theoretical point to keep in mind. In the numerical part, the absence of visible bias is checked empirically by comparison with a long reference MCMC chain.

Choice of the Polynomial P

The polynomial P controls the richness of the control-variate dictionary. A higher degree gives more control variates and may reduce the variance more strongly, but it also increases the dimension of the regression problem.

For a d -dimensional parameter x , the number of non-constant polynomial terms of degree at most p is

$$m(d, p) = \binom{d+p}{d} - 1. \quad (1.37)$$

In our GARCH(1,1) case, $d = 3$.

Degree 1. For a first-order polynomial,

$$P(x) = a^\top x, \quad a \in \mathbb{R}^d.$$

Then

$$\nabla P(x) = a, \quad \Delta P(x) = 0.$$

Thus, (1.35) becomes

$$\tilde{f}(x) = f(x) + a^\top z(x). \quad (1.38)$$

The variance to minimise is

$$\text{Var}_\pi(f + a^\top z).$$

The optimal coefficient is

$$a^* = -\Sigma_{zz}^{-1}\sigma(z, f), \quad \Sigma_{zz} = \text{Cov}_\pi(z, z), \quad \sigma(z, f) = \text{Cov}_\pi(z, f)$$

Since the expectations are unavailable in closed form, they are estimated from the MCMC sample.

Let

$$z_i = z(\omega^{(i)}), \quad f_i = f(\omega^{(i)}).$$

The empirical covariance estimates are

$$\hat{\Sigma}_{zz} = \frac{1}{N} \sum_{i=1}^N (z_i - \bar{z})(z_i - \bar{z})^\top,$$

and

$$\hat{\sigma}(z, f) = \frac{1}{N} \sum_{i=1}^N (z_i - \bar{z})(f_i - \bar{f}).$$

Therefore,

$$\hat{a} = -\hat{\Sigma}_{zz}^{-1}\hat{\sigma}(z, f).$$

Link with OLS. Consider the linear regression

$$f_i = \beta_0 + \beta^\top z_i + \varepsilon_i. \tag{1.39}$$

The OLS slope is

$$\hat{\beta} = \hat{\Sigma}_{zz}^{-1}\hat{\sigma}(z, f).$$

Thus,

$$\hat{a} = -\hat{\beta}.$$

The degree-1 Zero-Variance estimator is therefore

$$\boxed{\hat{\mu}_{\text{ZV},1} = \bar{f} - \hat{\beta}^\top \bar{z}.} \tag{1.40}$$

This establishes the key link:

$$\boxed{\text{Zero-Variance coefficient} = -\text{OLS slope}.}$$

For $d = 3$, degree 1 produces

$$m(3, 1) = \binom{4}{3} - 1 = 3$$

control variates.

Degree 2. For a quadratic polynomial,

$$P(x) = a^\top x + \frac{1}{2}x^\top Bx, \quad B = B^\top.$$

Then

$$\nabla P(x) = a + Bx, \quad \Delta P(x) = \text{tr}(B).$$

Hence,

$$\tilde{f}(x) = f(x) - \frac{1}{2}\text{tr}(B) + (a + Bx)^\top z(x). \tag{1.41}$$

Expanding the quadratic correction gives the following regressors:

- the d first-order controls:

$$z_j(x), \quad j = 1, \dots, d;$$

- the d diagonal quadratic controls:

$$x_j z_j(x) - \frac{1}{2}, \quad j = 1, \dots, d;$$

- the $d(d-1)/2$ off-diagonal quadratic controls:

$$x_i z_j(x) + x_j z_i(x), \quad 1 \leq i < j \leq d.$$

The total number of degree-2 controls is

$$d + d + \frac{d(d-1)}{2} = \frac{d(d+3)}{2}.$$

For $d = 3$,

$$m(3, 2) = \frac{3(3+3)}{2} = 9.$$

Thus, degree 2 enriches the control space from 3 to 9 regressors.

General Regression Form of the ZV Estimator

For a chosen polynomial degree, all control terms are collected in a matrix

$$W = \begin{pmatrix} W_1^\top \\ \vdots \\ W_N^\top \end{pmatrix},$$

where W_i is the vector of control variates evaluated at $\omega^{(i)}$. The corrected estimator can be written in the generic form

$$\tilde{f}_i = f_i - \widehat{\beta}^\top W_i,$$

where $\widehat{\beta}$ is obtained by regressing f_i on W_i .

With an intercept in the regression, the final estimator is

$$\boxed{\widehat{\mu}_{ZV} = \bar{f} - \widehat{\beta}^\top \bar{W}.} \quad (1.42)$$

This formula is the practical version used in the numerical implementation.

The Three ZV Regression Estimators

In the numerical experiments, three estimators are compared.

ZV-OLS. Ordinary least squares is used to estimate the regression coefficients:

$$\widehat{\beta}^{OLS} = \arg \min_{\beta} \|f_c - W_c \beta\|_2^2,$$

where $f_c = f - \bar{f}$ and $W_c = W - \bar{W}$. This is the direct regression implementation of the Zero-Variance method.

ZV-Lasso. Lasso regression adds an ℓ_1 -penalty:

$$\widehat{\beta}^{Lasso} = \arg \min_{\beta} \left\{ \frac{1}{2N} \|f_c - W_c \beta\|_2^2 + \lambda \|\beta\|_1 \right\}.$$

The penalty encourages sparsity and stabilises the regression when the number of control variates becomes large or when the columns of W are collinear.

ZV-Post-Lasso. First, Lasso selects an active set

$$\widehat{S} = \{j : \widehat{\beta}_j^{Lasso} \neq 0\}.$$

Then OLS is refitted only on the selected controls:

$$\widehat{\beta}^{Post} = \arg \min_{\beta_{\widehat{S}}} \|f_c - W_{c,\widehat{S}}\beta_{\widehat{S}}\|_2^2.$$

This reduces the shrinkage bias introduced by Lasso while keeping the selected lower-dimensional control space.

Summary. The Zero-Variance principle transforms the problem of variance reduction into a regression problem. The posterior score $z(\omega)$ provides the basic control variates, the polynomial P determines the size of the control dictionary, and OLS or Lasso-type methods estimate the coefficients used to reduce the Monte Carlo variance.

Chapter 2

Question 1 — Metropolis–Hastings Sampler

This first question aims to construct and validate a Metropolis–Hastings sampler for the Bayesian GARCH(1,1) posterior introduced in the methodological framework. The objective is not yet to reduce variance, but to obtain reliable posterior samples that will later serve as the baseline for the Zero-Variance estimators of Questions 2 and 3. The validation is carried out in three steps: first on simulated data, then on real EUR/USD returns, and finally through a repeated-chain boxplot experiment designed to isolate Monte Carlo error.

2.1 Validation on Simulated Data

Simulation setup

A first validation is performed on simulated GARCH(1,1) data. We generate $T = 5,000$ returns using the true parameter

$$\omega^* = (0.05, 0.10, 0.85).$$

The persistence of the data-generating process is therefore

$$\omega_2^* + \omega_3^* = 0.95,$$

which implies slowly decaying volatility shocks and visible volatility clustering.

```
1 rng_data = np.random.default_rng(123)
2 omega_true = np.array([0.05, 0.10, 0.85])
3 T = 5000
4 r = simulate_garch(omega_true, T, rng_data)
5 prior = Prior.default()
6
7 chain_pilot, acc_pilot, Sigma_prop = adaptive_rwm(
8     r, prior, n_iter=3000,
9     omega0=np.array([0.1, 0.1, 0.7]),
10    rng=np.random.default_rng(99),
11    pilot_iter=2000,
12 )
```

Listing 2.1: Simulating GARCH data and running the adaptive RWM pilot.

The simulated returns and the corresponding conditional volatility are shown in Figure 2.1. The returns are centered around zero, while the volatility series exhibits persistent clusters of high and low volatility, which is the main stylized behavior captured by a GARCH model.

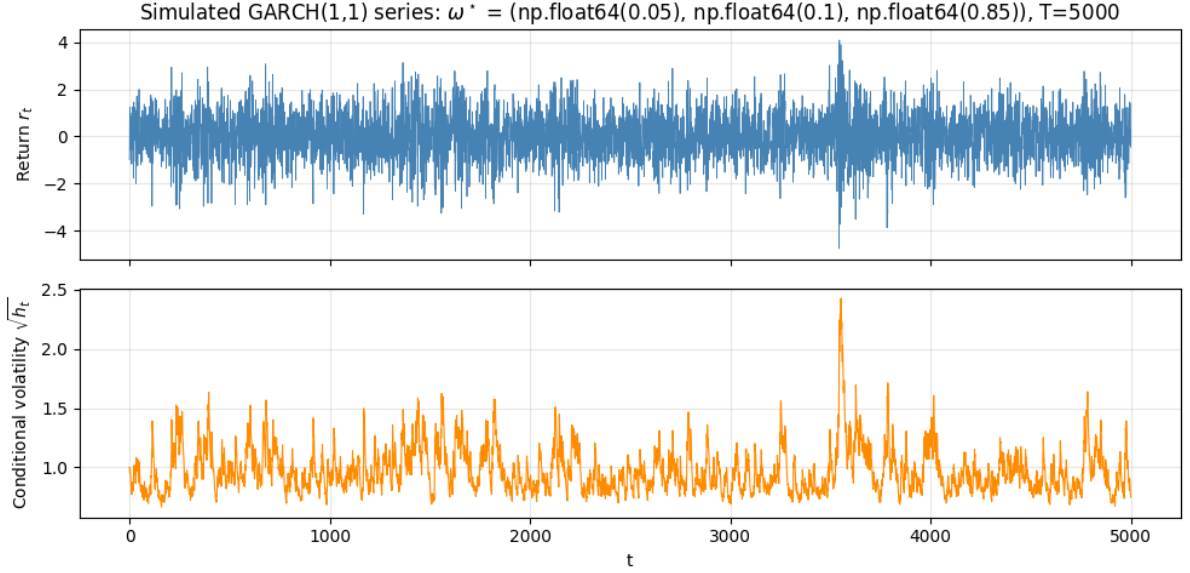


Figure 2.1: Simulated returns and conditional volatility.

Sampler calibration

The Random Walk Metropolis proposal covariance is calibrated using a preliminary pilot run. The Roberts–Gelman–Gilks scaling rule is used:

$$\Sigma_{\text{prop}} = \frac{2.38^2}{d} \hat{\Sigma}_{\pi}, \quad d = 3,$$

where $\hat{\Sigma}_{\pi}$ is estimated from the pilot chain.

The pilot acceptance rate is

$$\text{Pilot acceptance rate} = 0.401.$$

This value indicates that the proposal scale is well calibrated for this three-dimensional posterior. A very small acceptance rate would mean that most proposals are rejected, while a very high rate would usually indicate that the proposal steps are too small.

Reference chain

After calibration, a long reference chain is run with $N = 300,000$ iterations and a burn-in of 5,000 iterations. The goal of this chain is to obtain a stable approximation of the posterior distribution and to serve as a benchmark for the later variance-reduction comparisons.

The main-chain starting point is

$$\omega_0 = (0.05504171, 0.09970520, 0.84911843).$$

The reference chain gives:

$$\text{Runtime} = 5.8 \text{ s}, \quad \text{Acceptance rate} = 0.400.$$

The posterior summaries obtained from this long chain are:

$$\hat{\mathbb{E}}[\omega \mid r] = (0.06672617, 0.09716468, 0.83450479),$$

$$\hat{\text{sd}}(\omega \mid r) = (0.01363984, 0.01192907, 0.02245623),$$

and

$$ESS(\omega_1, \omega_2, \omega_3) = (14068, 15337, 12290).$$

The estimated persistence is

$$\widehat{\omega_2 + \omega_3} = 0.09716468 + 0.83450479 = 0.93166947.$$

Since this value is below one, the posterior mass remains inside the stationary region.

Trace plots and posterior densities

Figure 2.2 presents two complementary diagnostics. The trace plots show that the chain fluctuates around stable regions without visible trend or long flat sections. The posterior densities are unimodal and roughly Gaussian, which supports the use of a Gaussian random-walk proposal calibrated by an empirical covariance matrix.

The red dashed lines represent the posterior means estimated from the long chain, while the green dotted lines represent the true data-generating parameters. The two are close, but not exactly identical. This difference is expected because the sample size is finite: the posterior is concentrated near the true value, but not exactly centered on it.

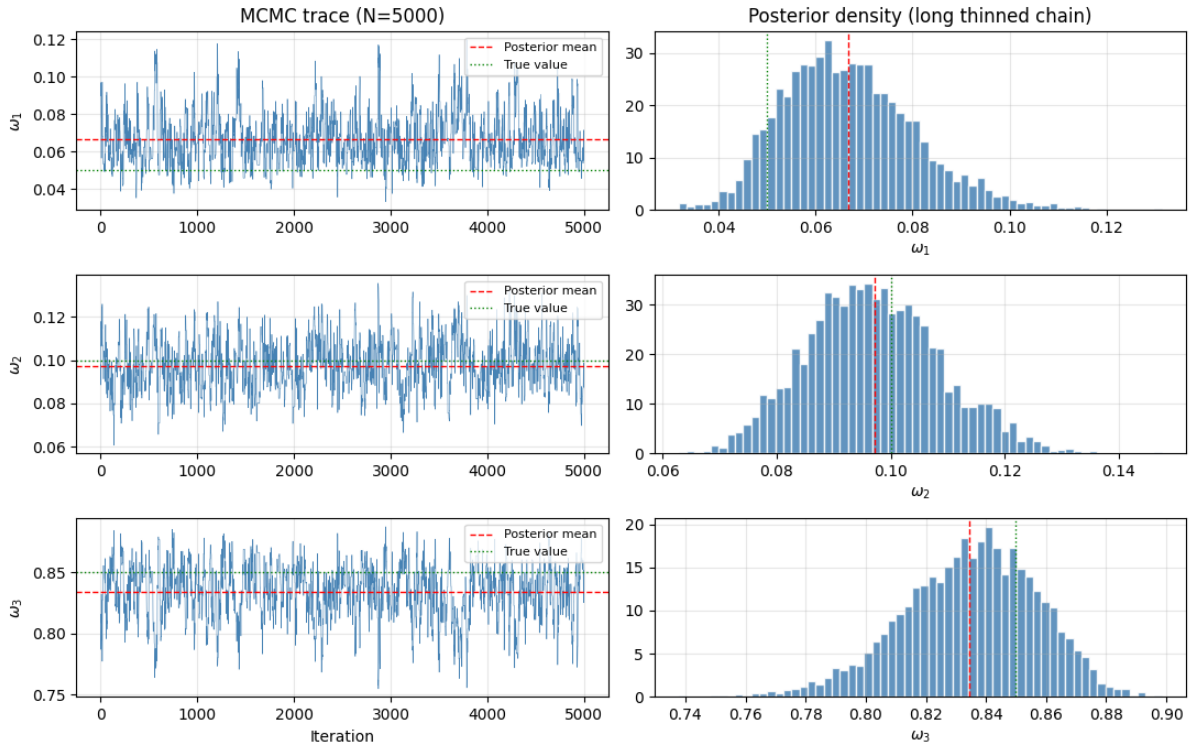


Figure 2.2: Trace plots and posterior densities on simulated data.

2.2 Application to Real EUR/USD Returns

Data preparation

The same sampler is then applied to daily EUR/USD log-returns from 2019-01-01 to 2024-12-31. The price series is downloaded using `yfinance`, and the log-returns are demeaned before estimation:

$$r_t = \log(S_t) - \log(S_{t-1}).$$

```
1 ticker = "EURUSD=X"
2 data = yf.download(
3     ticker, start="2019-01-01", end="2024-12-31",
4     progress=False, auto_adjust=True
5 )
6 prices = data["Close"].dropna().squeeze()
7 r_fx = np.log(prices).diff().dropna().to_numpy()
8 r_fx = r_fx - r_fx.mean()
```

Listing 2.2: Loading EUR/USD log-returns.

Sampler tuning on real data

For the real-data experiment, the first pilot run produced a very low acceptance rate:

$$\text{Pilot acceptance rate} = 0.023.$$

This indicates that the proposal steps were too large for the posterior geometry. After retuning the proposal scale, the main chain achieved:

$$\text{Main-chain acceptance rate} = 0.307.$$

This value is much more appropriate and indicates that the chain is able to move through the posterior region without being excessively rejected.

Filtered volatility

The posterior mean $\hat{\omega}$ is inserted into the GARCH recursion to obtain the filtered conditional volatility $\sqrt{\hat{h}_t}$. The result is shown in Figure 2.3. The volatility spikes correspond to periods of market turbulence, especially around 2020 and 2022. This provides a qualitative sanity check: the fitted GARCH model captures time-varying volatility in the exchange-rate returns.

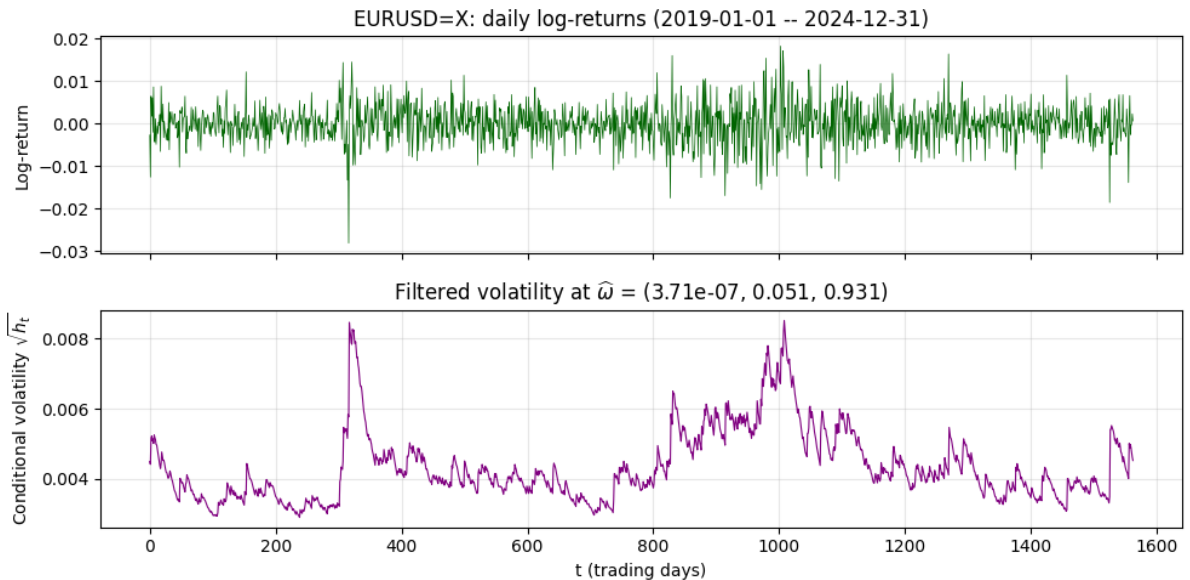


Figure 2.3: EUR/USD log-returns and filtered volatility.

2.3 Repeated-Chain Boxplot Experiment

Experimental design

To evaluate the Monte Carlo variability of the ordinary MCMC estimator, we run $n_{\text{rep}} = 30$ independent adaptive-RWM chains on the same fixed simulated dataset. Only the MCMC random seed changes across replications.

The fixed dataset is simulated with

$$T = 5,000, \quad \omega^* = (0.05, 0.08, 0.90).$$

Each chain has $N = 16,000$ post-burn-in iterations.

This design isolates Monte Carlo error: the dataset is fixed, so the spread of the estimates across replications is due only to the randomness of the MCMC algorithm, not to data-sampling variability.

```

1 n_replications = 30
2 posterior_means = []
3
4 for rep in range(n_replications):
5     rng_mcmc = np.random.default_rng(5000 + rep)
6     chain_rep, acc_rep, _ = adaptive_rwm(
7         r=returns_box,
8         prior=prior,
9         n_iter=n_iter_box,
10        omega0=omega0_box,
11        rng=rng_mcmc,
12        pilot_iter=2000,
13        burn_in=burn_in_box,
14    )
15    posterior_means.append(chain_rep.mean(axis=0))

```

Listing 2.3: Repeated independent MCMC runs on a fixed dataset.

Interpretation of the boxplots

Figure 2.4 reports the posterior-mean estimates across the 30 independent chains for the four targets:

$$\omega_1, \quad \omega_2, \quad \omega_3, \quad \omega_2 + \omega_3.$$

The width of each boxplot measures the Monte Carlo error of the ordinary MCMC estimator. The blue dotted line represents the average across replications, which is used as a proxy for the posterior mean. The red dashed line represents the data-generating value. These two lines are close but not necessarily identical, because the posterior mean may differ slightly from the true parameter in a finite sample.

These boxplots provide the baseline dispersion that the Zero-Variance estimators in Questions 2 and 3 aim to reduce.

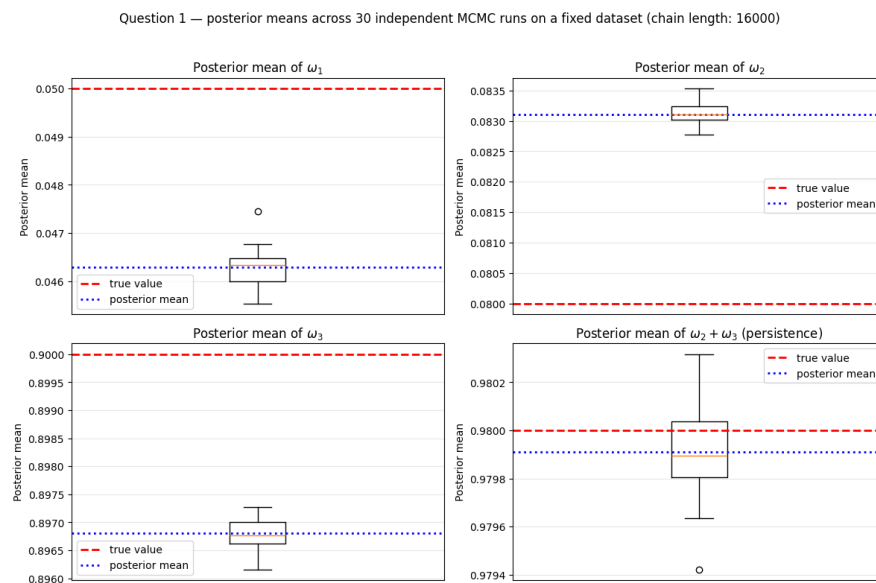


Figure 2.4: Posterior-mean estimates across 30 independent chains.

Conclusion of Question 1

The Metropolis–Hastings sampler was assessed on simulated GARCH(1,1) data and then applied to real EUR/USD returns. On simulated data, the acceptance rate around 0.40, the stable trace plots,

the unimodal posterior densities, and the effective sample sizes all indicate satisfactory sampling behavior. On real EUR/USD data, the retuned sampler achieved a reasonable acceptance rate of 0.307 and produced a filtered volatility series consistent with the observed clustering of financial returns.

Finally, the repeated-chain boxplot experiment provides a clear baseline for ordinary MCMC Monte Carlo error. This baseline will be used in the next questions to measure how much the Zero-Variance control variates reduce the dispersion of posterior mean estimators.

Chapter 3

Question 2 — First-Order Control Variates

This second question applies the Zero-Variance framework to the Bayesian GARCH(1,1) posterior using a first-order polynomial trial function. The goal is to reduce the Monte Carlo variability of posterior mean estimators without changing the target expectation. The ordinary MCMC estimator obtained in Question 1 is used as a baseline, and the first-order ZV estimator is compared against it through repeated independent runs.

3.1 First-Order ZV Construction

Target functions

We consider the four posterior quantities:

$$f_1(\omega) = \omega_1, \quad f_2(\omega) = \omega_2, \quad f_3(\omega) = \omega_3, \quad f_4(\omega) = \omega_2 + \omega_3.$$

The last target,

$$\omega_2 + \omega_3,$$

measures the persistence of volatility. Values close to one indicate slow mean-reversion of the conditional variance and persistent volatility clustering. The stationarity constraint requires

$$\omega_2 + \omega_3 < 1.$$

Degree-1 polynomial

For Question 2, we use a degree-1 polynomial:

$$P(\omega) = a^\top \omega.$$

Therefore,

$$\nabla P(\omega) = a, \quad \Delta P(\omega) = 0.$$

The Zero-Variance corrected function becomes:

$$\tilde{f}(\omega) = f(\omega) + a^\top z(\omega),$$

where

$$z(\omega) = -\frac{1}{2} \nabla \log \tilde{\pi}(\omega).$$

Since

$$\pi(\omega \mid r) = \frac{\tilde{\pi}(\omega)}{Z},$$

we have

$$\nabla \log \pi(\omega \mid r) = \nabla \log \tilde{\pi}(\omega),$$

because the normalising constant Z does not depend on ω . Thus, the vector $z(\omega)$ can be computed directly from the unnormalised log-posterior.

3.2 Regression-Based Implementation

OLS interpretation

For a fixed MCMC sample

$$(\omega^{(i)})_{i=1}^N,$$

we compute the posterior-score controls:

$$z_i = z(\omega^{(i)}), \quad i = 1, \dots, N.$$

For degree 1, the control matrix has three columns:

$$W_i = z_i \in \mathbb{R}^3.$$

The optimal coefficient a can be estimated through a linear regression of the target values $f(\omega^{(i)})$ on the control matrix W . With the regression convention used in the implementation,

$$f_i = \beta_0 + \beta^\top W_i + \varepsilon_i,$$

and the ZV estimator is written as:

$$\hat{\mu}_{\text{ZV}} = \bar{f} - \hat{\beta}^\top \bar{W}.$$

This formula corresponds to using the coefficient

$$\hat{a} = -\hat{\beta}$$

in the correction

$$\tilde{f} = f + a^\top z.$$

Algorithm

The computational protocol is as follows:

1. Run a posterior MCMC chain on the fixed dataset.
2. Compute the posterior-score vector for each retained state (`all_z`).

$$z(\omega^{(i)}) = -\frac{1}{2} \nabla \log \tilde{\pi}(\omega^{(i)})$$

3. Build the degree-1 control matrix W with three columns (`build_controls`).
4. Regress each target $f(\omega)$ on W using OLS to obtain $\hat{a}$ (`zv_estimator_ols`).
5. Compute the corrected estimator

$$\hat{\mu}_{\text{ZV}} = \bar{f} - \hat{\beta}^\top \bar{W}.$$

```
1 Z_q2 = all_z(chain_q2, r_q2, prior)
2 W1_q2 = build_controls(chain_q2, Z_q2, degree=1)
3
4 for name, f_vals in f_dict_q2.items():
5     mu_mc = float(np.mean(f_vals))
6     mu_zv, residuals = zv_estimator_ols(f_vals, W1_q2)
7
8     var_mc = float(np.var(f_vals, ddof=1))
9     var_res = float(np.var(residuals, ddof=1))
10    vr_factor = var_mc / var_res
```

Listing 3.1: Degree-1 ZV-OLS estimation.

3.3 Single-Run Variance Reduction

For a first diagnostic, the variance reduction is measured along one MCMC chain using:

$$\text{VRF} = \frac{\text{Var}(f(\omega^{(i)}))}{\text{Var}(\text{residuals})}.$$

A value greater than one means that the control variates reduce the variability of the target function along the chain. In this experiment, the degree-1 controls already provide substantial variance reduction because the posterior score $z(\omega)$ is strongly correlated with the target quantities.

3.4 Repeated-Run Experiment

Reference posterior mean

Before comparing repeated runs, we run a **long reference chain** ($N_{\text{ref}} = 200,000$ iterations) on the fixed dataset. We treat this average as a quasi-exact reference $\mathbb{E}_{\pi}[f]$. This mirrors Mira, Solgi & Imparato (2013), who use a chain of 10^7 iterations as a reference.

Experimental design

We run $n_{\text{rep}} = 30$ independent chains ($N = 20,000$) on the **same fixed dataset**, varying only the MCMC seed. This design isolates Monte Carlo error: the dataset is fixed, so the spread across runs comes from the sampling algorithm, not from data variability.

For each replication and for each target f_j , two estimators are recorded:

$$\hat{\mu}_{\text{MCMC}} = \frac{1}{N} \sum_{i=1}^N f_j(\omega^{(i)}) \quad \hat{\mu}_{\text{ZV}} = \bar{f}_j - \hat{\beta}_j^{\top} \bar{W}.$$

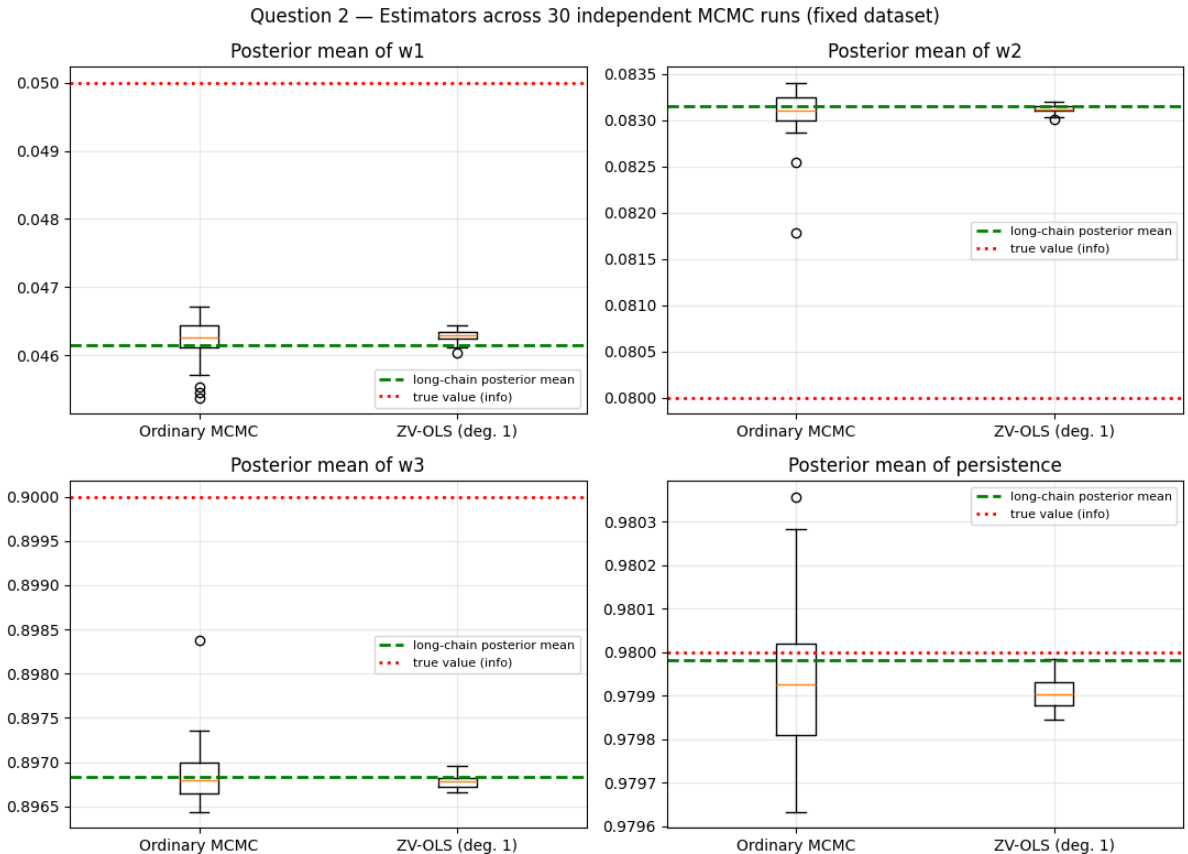


Figure 3.1: MCMC and degree-1 ZV-OLS estimates across repeated runs.

Interpretation

Figure 3.1 compares the ordinary MCMC estimates with the degree-1 ZV-OLS estimates across the 30 replications. The ZV-OLS boxes are visibly narrower than the ordinary MCMC boxes, showing a clear reduction in Monte Carlo dispersion.

The variance-reduction factors obtained in the experiment are approximately:

Target	Variance-reduction factor
ω_1	13.5
ω_2	49.9
ω_3	24.8
$\omega_2 + \omega_3$	18.6

These values show that even a first-order polynomial can substantially improve posterior mean estimation. The largest gain is obtained for ω_2 , suggesting that this target is particularly well explained by the first-order posterior-score controls.

The boxplots remain centered around the long-chain reference value. This is important: the ZV correction reduces variance without visibly shifting the target expectation. The red line corresponding to the true parameter is close to the reference value because the posterior is well concentrated for $T = 5,000$, but the estimator is designed to estimate the posterior mean, not directly the data-generating value.

Conclusion of Question 2

The first-order Zero-Variance control variates provide a strong improvement over the ordinary MCMC estimator. By regressing each target function on the posterior-score controls, the estimator keeps the same posterior expectation while substantially reducing Monte Carlo variability. The repeated-run experiment confirms variance-reduction factors between roughly 10 and 50, depending on the target.

This confirms that the ZV principle is effective even with a simple degree-1 polynomial. The next question extends the control dictionary to higher-order polynomials and investigates how Lasso-based selection can be used when the number of control variates increases.

Chapter 4

Question 3 — Larger Control-Variate Spaces

This question extends the Zero-Variance approach by increasing the size of the control-variate dictionary. After the degree-1 construction of Question 2, we now use degree-2 polynomial controls and compare three regression strategies: ZV-OLS, ZV-Lasso and ZV-Post-Lasso. The objective is to study how a richer control space improves variance reduction, and when regularisation becomes useful.

4.1 Motivation

The degree-1 ZV estimator uses only the three components of the posterior-score vector $z(\omega)$. This already reduces the Monte Carlo variance, but the control space can be enriched by using a higher-degree polynomial P . For a quadratic trial function,

$$P(\omega) = a^\top \omega + \frac{1}{2} \omega^\top B \omega, \quad B = B^\top,$$

the ZV correction becomes

$$\tilde{f}(\omega) = f(\omega) - \frac{1}{2} \text{tr}(B) + (a + B\omega)^\top z(\omega).$$

This construction introduces additional controls involving products between the parameters and the posterior score. These controls are more expressive than the degree-1 controls and can therefore lead to much stronger variance reduction.

However, when the number of control variates becomes large, ordinary least squares may become expensive or unstable. If $W \in \mathbb{R}^{N \times p}$ is the control matrix, a direct OLS fit requires forming $W^\top W$ and solving a $p \times p$ linear system, with computational cost approximately

$$O(Np^2 + p^3).$$

Moreover, the columns of W may be highly collinear. This motivates the use of Lasso-based methods to stabilise the regression and select useful controls.

4.2 Degree-2 Control Matrix

For degree 1, the control matrix contains the three components of z :

$$z_1(\omega), \quad z_2(\omega), \quad z_3(\omega).$$

For degree 2, the canonical ZV basis adds the following terms:

- the three diagonal quadratic controls

$$\omega_i z_i(\omega) - \frac{1}{2}, \quad i = 1, 2, 3;$$

- the three symmetric off-diagonal controls

$$\omega_i z_j(\omega) + \omega_j z_i(\omega), \quad i < j.$$

Thus, for $d = 3$, the total number of controls is

$$d + d + \frac{d(d-1)}{2} = \frac{d(d+3)}{2} = 9.$$

matching the formula of Mira et al. (2013).

```

1 Z_q3 = all_z(chain_q3, r_q3, prior)
2
3 W1_q3 = build_controls(chain_q3, Z_q3, degree=1) # shape (N, 3)
4 W2_q3 = build_controls(chain_q3, Z_q3, degree=2) # shape (N, 9)

```

Listing 4.1: Building degree-1 and degree-2 control matrices.

4.3 Regression Estimators

For each target function

$$f(\omega) \in \{\omega_1, \omega_2, \omega_3, \omega_2 + \omega_3\},$$

we compare the following estimators on the same degree-2 control matrix.

```

1 for name, f_vals in f_dict_q3.items():
2     mu_ols, residuals_ols = zv_estimator_ols(f_vals, W2_q3)
3     mu_lasso = zv_estimator_lasso(f_vals, W2_q3, cv=3)
4     mu_post = zv_estimator_post_lasso(f_vals, W2_q3, cv=3)

```

Listing 4.2: Single-run comparison of degree-2 ZV estimators.

4.4 Repeated-Run Experiment

Experimental design

As in Question 2, we use the same fixed simulated dataset and run $n_{\text{rep}} = 30$ independent MCMC chains with $N = 20,000$ iterations each. Only the MCMC seed changes across replications. This design isolates the Monte Carlo variability of the estimators.

For each replication and each target, four estimators are recorded:

$$\hat{\mu}_{MCMC}, \quad \hat{\mu}_{ZV-OLS}, \quad \hat{\mu}_{ZV-Lasso}, \quad \hat{\mu}_{ZV-Post}.$$

Interpretation

Figure 4.1 shows that all ZV estimators strongly reduce the spread of the ordinary MCMC estimates. Moving from degree 1 to degree 2 provides a major improvement: the boxplots become much narrower, meaning that the Monte Carlo error is sharply reduced.

In the experiment, the variance-reduction factors for ZV-OLS and ZV-Post-Lasso are of the order of several hundred to several thousand, depending on the target. This is consistent with the behavior reported by Mira et al. for higher-order ZV controls.

For the current GARCH(1,1) problem, the degree-2 dictionary has only 9 columns. Therefore, full OLS remains stable and generally performs as well as, or better than, Lasso-based alternatives. Lasso becomes more useful when the dictionary is much larger, for example with higher-degree controls or higher-dimensional models.

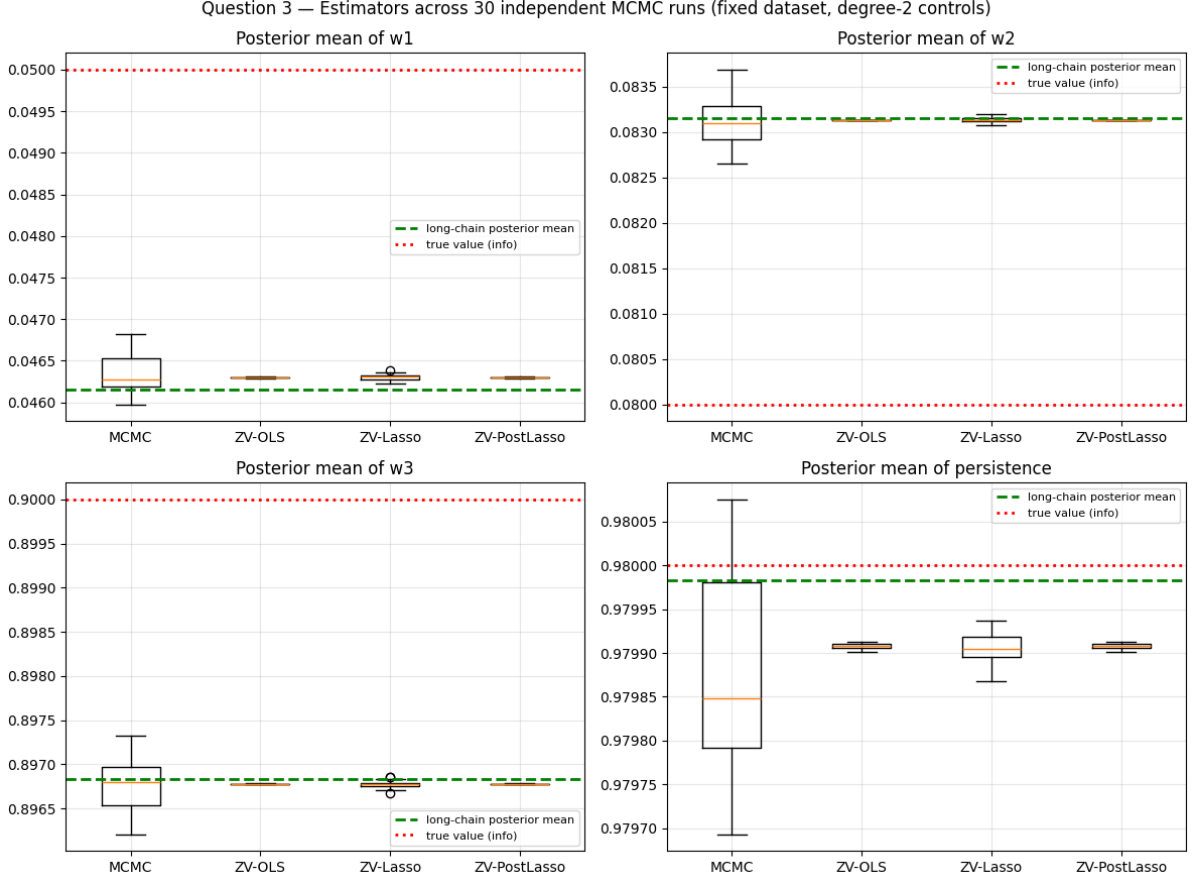


Figure 4.1: Repeated-run comparison of MCMC and degree-2 ZV estimators.

Bias and centering

The corrected estimators remain centered around the long-chain posterior mean, so no visible bias is detected at the scale of this repeated-run experiment. Nevertheless, exact unbiasedness should not be claimed only from the formal ZV identity, because the posterior is restricted to the admissible region $\mathcal{A}$ and boundary terms may remain.

In addition, the same chain is used both to estimate the regression coefficients and to compute the corrected average. This creates a finite-sample dependence between the fitted correction and the estimator. This motivates the cross-fitting check presented next.

4.5 Cross-Fitting Check

The standard ZV-OLS estimator fits the regression coefficients and computes the corrected average using the same MCMC sample. To reduce this finite-sample dependence, we use a cross-fitting strategy.

The chain is split into two blocks, A and B . The coefficient $\hat{\beta}_A$ is fitted on block A and applied to block B , while $\hat{\beta}_B$ is fitted on block B and applied to block A . The cross-fitted estimator is then

$$\hat{\mu}_{CF} = \frac{1}{2} (\hat{\mu}^{(A)} + \hat{\mu}^{(B)}) .$$

The two blocks are not fully independent because they come from the same Markov chain, but cross-fitting avoids using each observation to correct itself. This is especially useful when the sample size is small, the chain is highly autocorrelated, or the control dictionary is large.

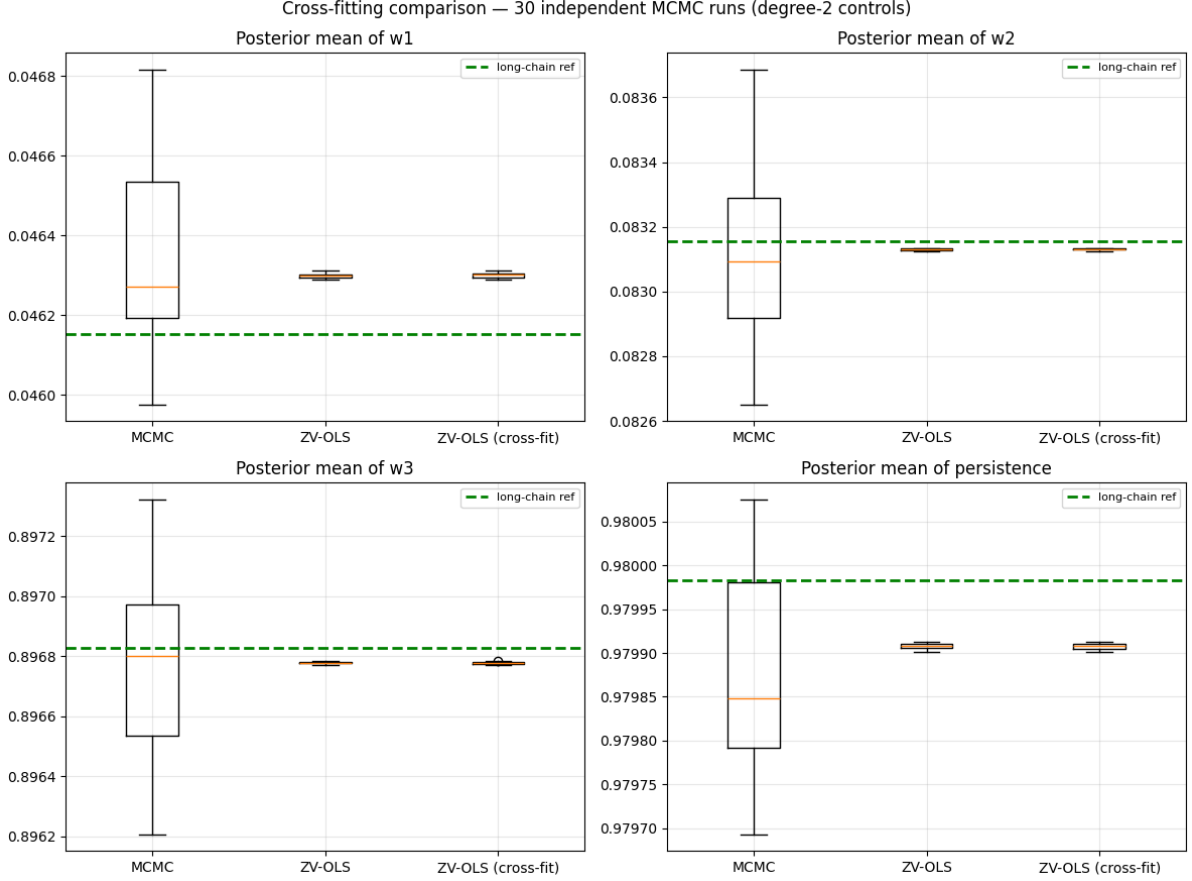


Figure 4.2: Comparison between standard ZV-OLS and cross-fitted ZV-OLS.

In our setting, the difference between the standard and cross-fitted estimators is limited. This is expected because the sample size is large and the number of degree-2 controls is moderate. Therefore, the benefit of cross-fitting is mainly diagnostic here.

4.6 Application to EUR/USD Log-Returns

The degree-2 ZV methodology is finally applied to the EUR/USD log-returns introduced in Question 1. This real-data experiment requires a specific calibration of the MCMC proposal.

On the simulated dataset, ω_1 is of order 10^{-2} . On EUR/USD returns, however, the posterior mean of ω_1 is much smaller, typically of order 10^{-6} . Therefore, a proposal scale calibrated for simulated data is too large in the ω_1 direction. Without retuning, the chain barely moves, and the ZV controls become numerically unstable.

Remark 4.1. *Zero-Variance correction is a post-processing variance-reduction method. It cannot compensate for a poorly mixed MCMC chain. A reasonably well-calibrated posterior sample is required before applying ZV controls.*

Interpretation of the EUR/USD results

Figure 4.3 shows that, after proper MCMC calibration, ZV-OLS substantially narrows the boxplots relative to ordinary MCMC. No visible bias is observed at the scale of the repeated-run experiment. The variance-reduction factors are smaller than those obtained on simulated data. They are typically between 80 and 350. This difference is explained by two main reasons:

- the EUR/USD posterior is much more concentrated, so the absolute MCMC noise is already small;

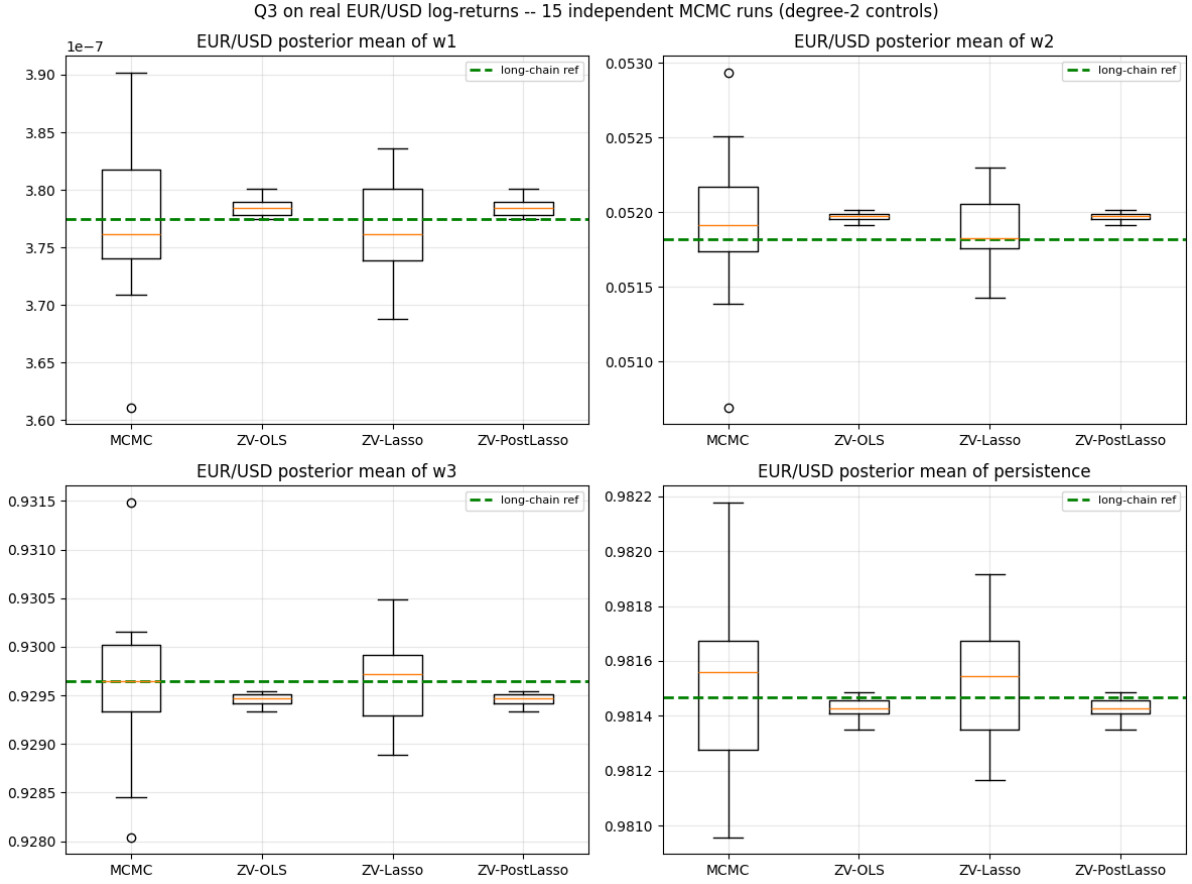


Figure 4.3: Degree-2 ZV variance reduction on EUR/USD returns.

- ω_1 is very close to the boundary $\omega_1 > 0$, which can make boundary effects more relevant in the integration-by-parts argument.

Thus, the real-data experiment confirms the practical usefulness of ZV corrections, while also showing that the magnitude of variance reduction depends on the posterior geometry and on the quality of MCMC calibration.

Conclusion of Question 3

Question 3 shows that enlarging the control-variate space from degree 1 to degree 2 leads to a substantial additional reduction in Monte Carlo error. On the simulated GARCH dataset, degree-2 ZV-OLS and ZV-Post-Lasso produce very narrow boxplots and variance-reduction factors of several hundred to several thousand.

In the present low-dimensional GARCH(1,1) setting, the degree-2 control matrix contains only 9 columns, so full OLS remains feasible and highly effective. Lasso and Post-Lasso are nevertheless important methodological extensions, because they become useful when the control dictionary grows and collinearity or computational cost becomes problematic.

The cross-fitting check confirms that using the same chain for fitting and correction does not create visible bias in the current experiment. Finally, the EUR/USD application shows that the method remains effective on real financial data, provided that the underlying MCMC sampler is properly calibrated.

Chapter 5

Question 4 (Bonus) — Regression on an MCMC Chain

The previous questions used linear regression to estimate the coefficients of the Zero-Variance control variates. In standard Monte Carlo, this regression is usually justified under independent samples. In our setting, however, the observations come from a Markov chain, and successive draws are autocorrelated. This bonus question studies whether this dependence affects the practical use of the regression-based ZV estimator.

5.1 Issue

The degree-2 ZV-OLS estimator relies on the regression coefficient

$$a^{\star} = -\Sigma_{zz}^{-1} \text{Cov}_{\pi}(z, f).$$

In practice, Σ_{zz} and $\text{Cov}_{\pi}(z, f)$ are estimated from the MCMC sample. Although the draws are not independent, the ergodic theorem still implies that empirical averages converge to their posterior expectations. Therefore, the estimated coefficient remains consistent under standard ergodicity assumptions.

The main issue is not consistency, but finite-sample variability. Because the chain is autocorrelated, the effective amount of information is smaller than the raw number of iterations N . This can inflate the variance of the estimated regression coefficient $\hat{a}$, and it invalidates naive OLS standard errors.

Possible remedies

Two simple strategies can be used to reduce the dependence in the regression sample:

Thinning

Keep one draw every k iterations in order to regress on less correlated observations. The ZV correction is then evaluated on the full chain.

Block averaging

Replace consecutive blocks of B MCMC draws by their averages. If B is large compared with the integrated autocorrelation time, block means are more weakly dependent than raw draws.

Both approaches reduce autocorrelation, but they also reduce the number of observations used to estimate the regression coefficient. This creates a trade-off between dependence reduction and information loss.

5.2 Empirical Autocorrelation

We first examine the autocorrelation of a representative MCMC chain with $N = 20,000$ iterations. For each parameter coordinate

$$\omega_1, \quad \omega_2, \quad \omega_3,$$

we compute the effective sample size and the integrated autocorrelation time:

$$\tau_k = \frac{N}{ESS_k}.$$

A large autocorrelation time means that many successive draws contain redundant information. Conversely, a large effective sample size indicates that the chain contains a substantial amount of usable information for posterior estimation.

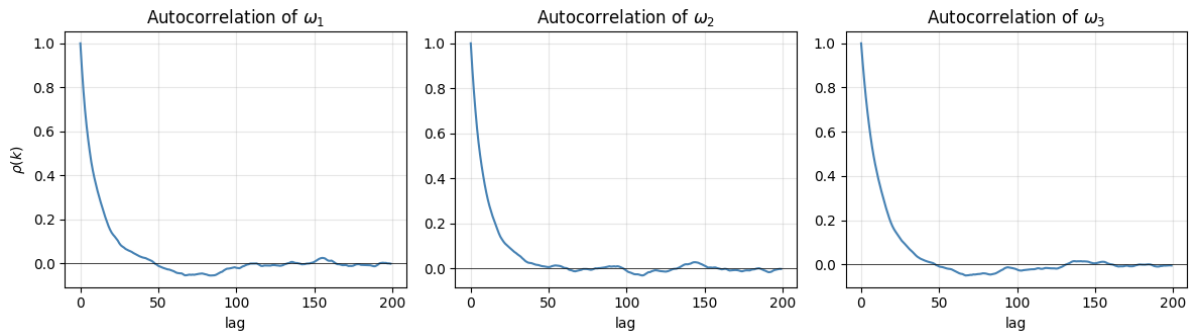


Figure 5.1: Autocorrelation functions and effective sample sizes.

Figure 5.1 is used to assess whether the dependence in the chain is strong enough to justify thinning or block averaging before fitting the ZV regression. In this experiment, the autocorrelation is present, as expected for a Random Walk Metropolis sampler, but the effective sample sizes remain large enough for stable coefficient estimation.

5.3 Comparison of Dependence-Reduction Strategies

Experimental design

We run $n_{\text{rep}} = 50$ independent MCMC chains, each with $N = 10,000$ iterations. The same simulated dataset is used in every replication, so the spread across runs measures Monte Carlo error only.

Seven strategies are compared:

raw chain, thinning $k \in \{5, 10, 20\}$, block averaging $B \in \{10, 50, 100\}$.

The raw strategy uses all MCMC draws to estimate the ZV coefficients. The thinning strategy estimates the regression coefficients on a thinned subsample, but applies the correction to the full chain. The block-averaging strategy fits the regression on block averages.

```

1 def zv_ols_with_thinning(chain, Z, f_vals, thin=1):
2     W_full = build_controls(chain, Z, degree=2)
3
4     W_thin = W_full[::thin]
5     f_thin = f_vals[::thin]
6
7     # OLS is fitted on the thinned subset.
8     # The correction is then evaluated on the full chain.
9     ...
10
11 def zv_ols_with_block_averaging(chain, Z, f_vals, block_size=50):

```

```

12 W_full = build_controls(chain, Z, degree=2)
13
14 # Consecutive observations are grouped into blocks.
15 # OLS is fitted on block means.
16 ...

```

Listing 5.1: Thinning and block-averaging variants.

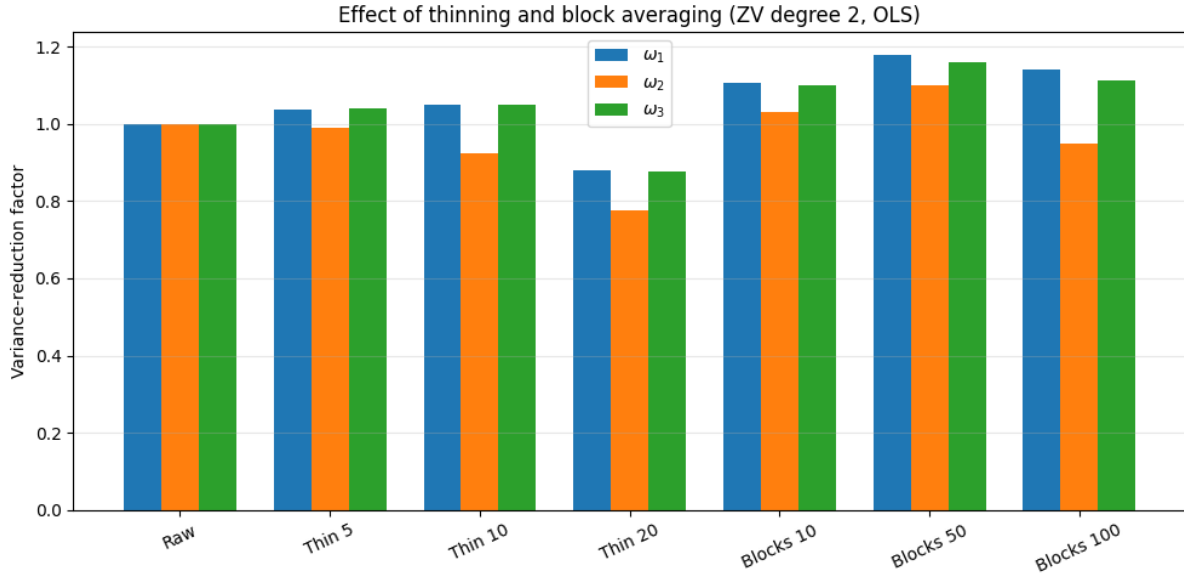


Figure 5.2: Raw chain, thinning and block-averaging comparisons.

5.3.1 Interpretation

Figure 5.2 compares the different dependence-reduction strategies relative to the raw-chain ZV estimator. The variance ratios are therefore normalised by the variance of the ZV estimator fitted on the full MCMC chain. Consequently, the value equal to one for the raw method is obtained by construction and should not be interpreted as the absence of variance reduction relative to ordinary MCMC.

The main conclusion is that thinning and block averaging do not provide a clear improvement over using the full MCMC chain. Although these strategies reduce the serial dependence of the observations used in the regression, they also reduce the amount of information available to estimate the control-variate coefficients. Thinning discards intermediate draws, while block averaging compresses the chain into a smaller number of averaged observations. This can increase the variability of the fitted regression coefficients and offset the potential benefit of lower autocorrelation.

This result is consistent with the role of the regression step in the Zero-Variance method. The coefficients are estimated from empirical posterior moments and covariances, which remain consistent under standard ergodicity assumptions even when the MCMC draws are correlated. Therefore, the main issue is not the validity of the regression estimator, but its finite-sample variability. In the present experiment, using the full post-burn-in chain provides the most stable implementation of the ZV correction.

Overall, the bonus experiment suggests that simple dependence-reduction strategies such as thinning or block averaging are not necessarily beneficial for regression-based control variates. For this GARCH application, it is preferable to keep all available MCMC draws and rely on the ergodic averages of the full chain rather than reducing the sample size before fitting the ZV regression.

Conclusion of Question 4

The bonus experiment confirms that the dependence of MCMC samples should be taken seriously, but it does not prevent the regression-based ZV estimator from working effectively. By ergodicity, the empirical covariance matrices used in the regression still converge to their posterior counterparts, so the estimated ZV coefficients remain consistent.

In the numerical comparison, thinning and block averaging did not improve much the variance reduction. Their main drawback is that they reduce the number of observations available for fitting the control-variate regression. In this experiment, the full raw chain provided the best trade-off between dependence and information content.

Therefore, the final methodological choice is to use the full post-burn-in chain for both coefficient estimation and the corrected ZV average, while relying on MCMC diagnostics such as ESS and autocorrelation plots to monitor the quality of the sample.

Conclusion

This project studied the use of Zero-Variance control variates to improve MCMC estimation in a Bayesian GARCH(1,1) framework. The results show that this approach can substantially reduce Monte Carlo error while preserving the same posterior target, provided that the underlying Markov chain is reasonably well mixed.

Summary of Results

Effectiveness of ZV.

The Zero-Variance approach proves highly effective for Bayesian estimation in the GARCH(1,1) model, with a moderate additional computational cost. Degree-1 controls already reduce Monte Carlo variance by factors around $10\text{--}50\times$, depending on the target. With the degree-2 polynomial basis (9 controls), the reduction reaches **several hundred to several thousand** on simulated data, in line with Mira et al. (2013), and about $80\times$ to $350\times$ on real EUR/USD log-returns. No visible bias is observed relative to the long-chain reference.

OLS vs Lasso vs Post-Lasso.

On the degree-2 basis, ZV-OLS remains the most efficient estimator because the number of controls is still small. Plain Lasso is less competitive, as coefficient shrinkage tends to under-correct the MCMC average. Post-Lasso gives a better compromise by selecting variables with Lasso and then refitting OLS without shrinkage. Regularisation becomes more relevant for larger and more collinear control dictionaries.

Cross-fitting.

Cross-fitting provides a principled way to reduce the finite-sample dependence between the estimated coefficients $\hat{\beta}$ and the corrected average. In the present experiments, where $N \geq 10^4$, its impact remains limited. It may nevertheless be useful for shorter chains, larger control spaces, or stronger autocorrelation.

Effect of autocorrelation (Q4).

Thinning and block averaging do not improve the variance reduction obtained with the full chain. They reduce the dependence between observations, but also discard useful information for estimating the regression coefficients. Overall, using the full MCMC chain is the most reliable strategy for the ZV point estimator in this study.

Overall, the experiments confirm that Zero-Variance control variates are a powerful post-processing tool for improving MCMC precision in constrained Bayesian time-series models. In this project, the best practical choice is to use a well-calibrated Random-Walk Metropolis sampler together with degree-2 ZV-OLS controls on the full chain.

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